

Key Takeaways

- Cloud computing is a model for enabling ubiquitous and convenient on-demand network access to a shared pool of configurable computing resources, that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- Cloud computing is a cost-effective, ~~seable~~ scalable, elastic, current, reliable, secure, metered, and self-serviced solution.
- Private, public and hybrid clouds are the three types of cloud models.
- Cloud services are of three types:
 - IaaS
 - PaaS
 - SaaS
- Manufacturing and retail industries have used cloud computing to help improve and scale their businesses.

Questions.

Q1. Important features of a public cloud?
Ans. i.) Metered - pricing
ii.) Self - serviced platform.

Q2. Cloud computing has evolved over time.

Ans. i.) Cloud computing.

Upcoming Info.

- List the benefits and challenges of migrating to cloud.
- Explain the five options of migrating to cloud.
- Describe the four steps of cloud migration plan.
- List the activities and resources that should move to cloud.
- Explain how the features of cloud can help preserve business continuity.
- Discover how cloud migration helps businesses.

Cloud Migration

1.) Cloud Adoption?

- i.) Greater Business Agility.
- ii.) Cost Savings.
- iii.) Reduced maintenance.
- iv.) Environment Friendly.

2.) Cloud Migration?

→ Cloud migration is the process of moving digital business operations into cloud.

→ Benefits :

- Increased Cost Savings
- Greater Flexibility
- Higher Scalability
- Increased flexibility
- Improved Performance
- Reduced Costs.

→ Challenges of Cloud Migration:

- Data Integrity
- Continued Operation
- Migrating Huge Data
- Interoperability and Portability
- Reliability and flexibility
- Downtime
- Security and Privacy
- Costs
- Lack of Resources
- Management of Cloud and Multi-Cloud Environment.

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Five Rs of Cloud Migration Strategy

- 1.) Rehost
- 2.) Refactor
- 3.) Revise
- 4.) Rebuild
- 5.) Replace

i.) Rehost:

- Business select an IaaS provider to rehost and reconstruct their application architecture on that infrastructure.
- The expected business outcome is no capital expenditure to get started, free up space in the data centres.
- Return on investment in cloud can be achieved rapidly in refactoring already existing code and frameworks are reused by the businesses that choose to refactor instead of IaaS.

11- Refactor:

- In refactoring already existing code and frameworks are reused by the businesses that choose to refactor. ~~instead~~ Instead of IAAS, they run their application on a PAAS platform as a service providers platform.
- The expected business outcome is updates can be rolled out quickly benefit from code portability greater cloud efficiency can be achieved in the areas of resources cost and speed.
- This strategy encompasses partial rewriting or expanding of the code base and then deploying it by either rehost or refactoring.
- The expected business outcome is application are scalable and agile new cloud capabilities can be adopted more easily. Different technology stacks can be used.

Revise

- The strategy encompasses partial rewriting or expanding of code base and then deploying it by either rehost or refactoring.
- The expected business outcome is applications are scalable and agile. new cloud capabilities can be adopted more easily.
- Different technology stacks can be used.

Rebuild

- Rebuild means rewriting and rearchitecting the application from scratch on a PaaS providers platform.
- This allows developers to leverage the advanced features offered by PaaS providers, but would be a labour intensive task.
- The expected business outcome is help drive innovation as your R and D teams can use different cloud services to develop the products.
- Application can be built faster, providing more time for testing and release activities.
- Operational costs can be reduced, giving you cost advantage as compared to on-premise, infrastructure.

Replace

- Enterprises can also opt to get rid of their old applications completely and switch to already available SaaS software as a service applications from different third party vendors.
 - The expected business outcome is standardize and follow industry best practices.
 - Business process driven approaches can be adopted quickly, improving collaboration in the team.
- Allows more investments into applications that create difference or provide competitive advantages so that you achieve greater ROI.

Cloud Migration Plan

• Four - Step Plan for Cloud Migration

- i.) Define strategy and perform assessment
- ii.) Plan for readiness and other processes.
- iii.) Adopt and migrate.
- iv.) Govern, manage, and optimize.

→ Define Strategy and Perform Assessment

1. Set business strategy
2. Use split approach for cloud implementation
3. Use tools to help you plan.

→ Set business strategy

- Establish clear business outcomes
- Define business justification.

→ Use split approach for cloud implementation.

- Perform a readiness assessment by:
- Identifying applications that need to move to cloud
- Choosing applications with minimum dependencies and assets.

→ Use tools to help you plan.

- Cloud Adoption Readiness Tool (CART) from Amazon Web Services (AWS).

- Azure Total Cost of Ownership (Tco) Calculator
- Azure Cost Management

→ Plan For Readiness and other Processes

- Address gaps uncovered in the assessment phase.
- Keep resources prepared with technical readiness
- Set in place processes to drive business and technology changes.
- Define a technology plan to enable business outcomes.
- Focus on building your baseline environment.

→ Adopt and Migrate

- Design, migrate and validate each application.
- Replicate existing resources to cloud with the least changes.
- Transition users from the existing solution to the cloud solution.

Preparation

- Set business objectives
- Prepare an estimate of the workloads to be migrated.
- Define the team structure, access requirements, and separation of

responsibilities.

- Specify processes and documentation required to review and approve changes.

Tools for Migration.

- AWS server Migration Service.
- AWS Database Migration Service.
- Azure Migrate.
- Google Cloud Platform (GCP) - Migrate for Compute Engine.



Govern, Manage and Optimize.

When an organization begins the journey to move to cloud, the final state of cloud adoption is not known, this final state starts appearing as and when your organization moves or deploys new applications to cloud. Considering the below aspects of managing and operating a cloud platform is important.

You should define governance solutions for your cloud environment that meet your organization's business requirements. help control risks and provide agility.

Some tools to help you.

- AWS management and governance service

- AWS control tower.
- AWS cloud watch
- Azure, Azure advisor, Azure monitor, Azure policy
- GCP security center alert center vault.

Business Continuity Through Cloud.

- 1) The first reason is best and development, new applications can be brought to the market faster by allowing teams to quickly create and deep provision, test and development environments. Development test environment can be scaled up and down quickly by leveraging cloud services.
2. Websites hosting as compared to traditional web hosting, running website on cloud using IAAS and PAAS can be less expensive storage, backup and recovery.
3. Storage, backup and recovery. Cloud computing helps in handling unpredictable demand and steadily growing storage needs. The capital expenditure for storage and complexity of managing data are avoided by organizations. planning and management of backup and recovery systems can be simplified.

4.) Web apps, cloud provides all the infrastructure required to deploy and support web apps including storage, web and application servers and networking resources. web apps can be quickly deployed and scaled as per the requirement.

5.) High-performance computing complex problems involving large amounts of data can be solved by leveraging high performance computing HPC on supercomputers, computer clusters or computer grids. Some of the use cases involving HPC are earthquake and protein folding simulations. Evaluating product designs, financial modeling and climate and weather predictions.

6.) Big data analysis, big data is a common term for huge datasets that contain potentially valuable trends patterns and associations. Cloud provides a huge amount of processing power required for mining datasets to locate or pull out hidden patterns.

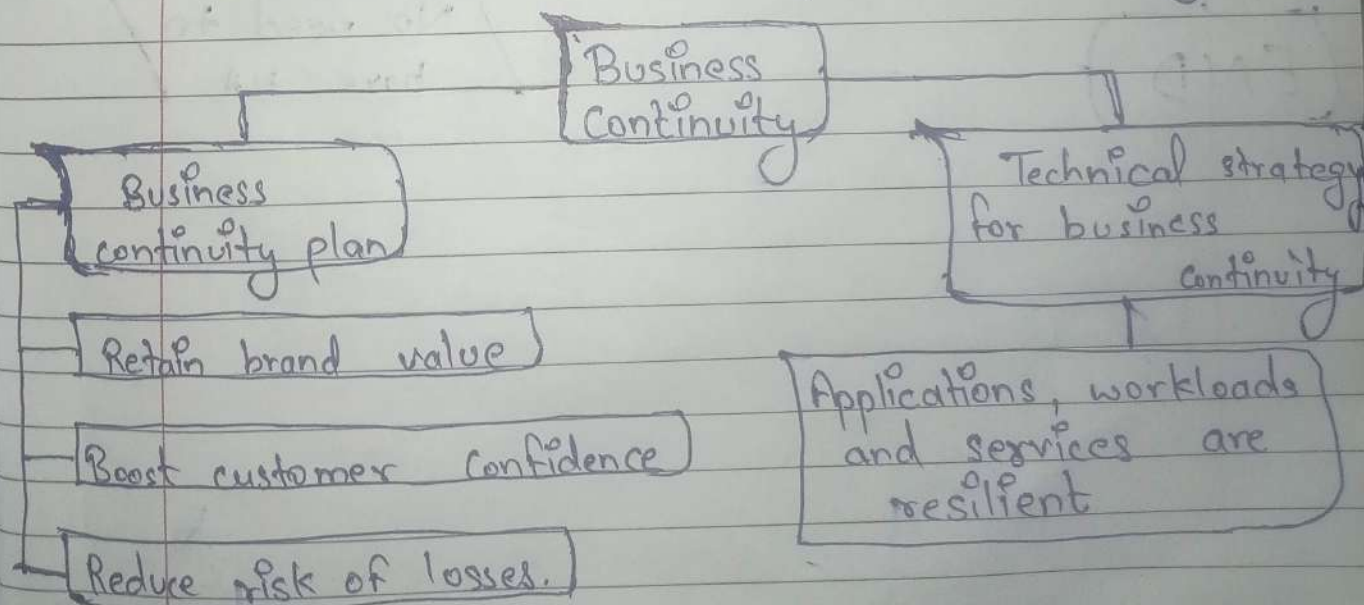
7. Extended datacenter cloud allows you to add capacity to your data center by adding virtual machines in cloud instead of incurring the costs of physical

adding hardware or space to your physical locations.

High Availability and Disaster Recovery

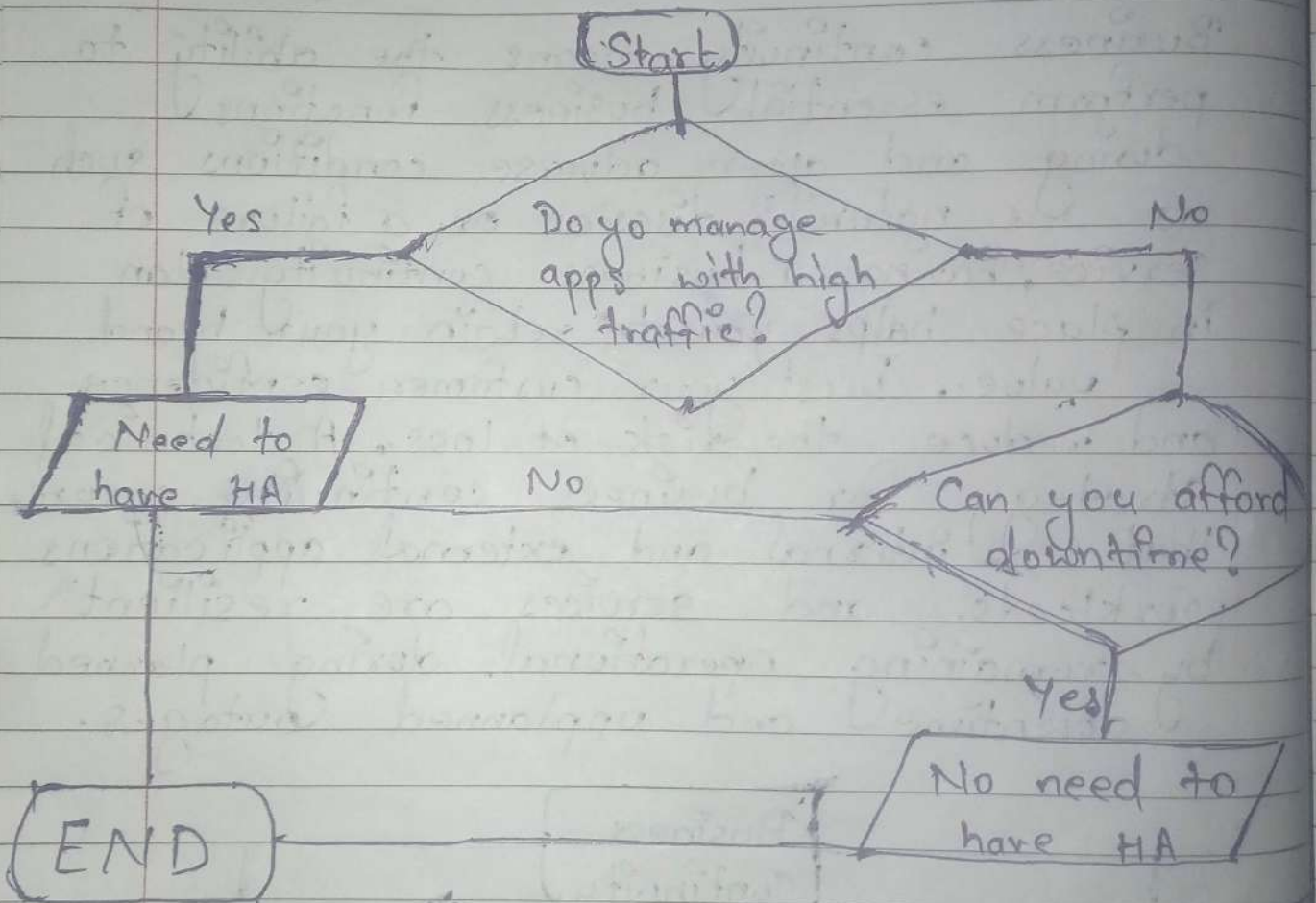
Preserving Business Continuity Through Cloud.

Business continuity means the ability to perform essential business functions during and after adverse conditions such as a natural disaster or a failure of service; having a business continuity plan in place helps you to retain your brand value, boost your customer confidence and reduce the risk of loss. A technical strategy for business continuity ensure that internal and external applications workloads, and services are resilient by remaining operational during planned downtime and unplanned outages.



High Availability

High availability (HA) is the ability of the application to continue running in a healthy state despite localized or transient failures.



Disaster Recovery

Disaster recovery (DR) is the process of recovering from major incidents, such as service disruption that affects an entire region.

- i.) Data backup
- ii.) Data archiving
- iii.) Using a backup to restore data.

Data Backup and Resiliency

- Data Replication and Backup
- Data replication allows systems to fail over quickly to a replica by copying data in near real time either synchronously or asynchronously. Data replication ensures that a replica of the data is readily available and can help reduce the time it takes to recover from an outage. Data replication should not be considered as a substitute for backups with data backups in place. You get quick access to files, you can be safe in case of power failures, be safe against hardware failures. Save money as you don't need physical hardware to back up data and applications.

Resiliency

Designed for resiliency in a cloud environment focuses on failures recovery rather than on avoiding failures. The goal is to return applications to a fully functioning state following a failure by responding to failures such that it avoid downtime or data loss.

In case of failure occurs. Then your team would be aware of whom to contact and start the recovery process. This makes it very simple for you to bring your application to life again, recovery time objective (RTO) Recovery time Object is the maximum acceptable time that an application can be unavailable after an incident. If the RTO is 70 minutes, the application must be restored to a running state within 70 minutes from the start of a disaster. You should consider a secondary deployment of application running in standby mode. If you have a very low RTO.

Recovery Point Objective.

Recovery point objective. RPO is the maximum duration of data loss that is acceptable during a disaster. If an inventory data is in a single database with no replication to other databases and performs backups every hours then you could lose data worth an hours.